



CHAMIKA ABEYKOON

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SUMMARY

Highly motivated **Software Engineering undergraduate** graduating in September 2026. Dedicated to crafting high-quality, impactful Web and Mobile applications, I specialize in turning complex problems into elegant, user-centric solutions. With a strong core philosophy centered on clean architecture, I actively pursue innovative solutions to technical challenges. Characterized by a collaborative mindset and an unwavering commitment to continuous learning, I am seeking a full-time position to leverage my modern tech stack, and hands-on project experience.

PROJECTS

AEGISHEALTH (FINAL YEAR PROJECT) | PYTHON, PYTORCH, FASTAPI, GRPC, NEXT.JS, ELECTRON.JS, SUPABASE | APRIL 2026

- Built an end-to-end privacy-preserving federated learning platform for cross-hospital health anomaly detection on non-IID data, keeping patient data local to each institution.
- Designed a dual-protocol backend with FastAPI (control plane) and TLS-secured gRPC (federated round coordination), including job orchestration, client registration, quorum handling, and model aggregation.
- Implemented LSTM-based time-series modeling with FedProx to improve convergence and robustness under heterogeneous client distributions.
- Integrated Differential Privacy and optional Homomorphic Encryption for secure aggregation and reduced inference leakage risk.
- Developed role-based server/client dashboards in Next.js and an Electron edge client for site-level participation, monitoring, audit visibility, and model release workflows

PHISHING SIMULATOR | NEXT.JS, REACT, NESTJS, TYPESCRIPT, SQLITE, TAILWIND CSS | SEPTEMBER 2025

- Developed a full-stack cybersecurity training platform featuring interactive email simulations, quiz-based learning, and progress tracking.
- Implemented role-based access control and secure JWT authentication.
- Integrated NLP analysis and real-time analytics to evaluate and improve user cybersecurity awareness.

FINFLOW-AI | ANDROID, KOTLIN, JETPACK COMPOSE, FIREBASE, KTOR, GEMINI API | AUGUST 2025

- Built an intelligent Android financial management application utilizing modern Android architecture and Material Design 3.
- Integrated the Gemini API to generate real-time, AI-powered financial insights and personalized coaching.
- Leveraged Kotlin Coroutines and Ktor for efficient asynchronous operations, backed by Firestore for real-time transaction tracking.

AI-POWERED CRM PLATFORM | NEXT.JS, NESTJS, TYPESCRIPT, MONGODB, OPENAI | MAY 2025

- Engineered a comprehensive Customer Relationship Management system featuring intelligent lead management and automated customer support.
- Developed role-based dashboards and integrated predictive analytics and sentiment analysis to drive sales automation.
- Implemented multi-factor authentication and custom AI chatbots to streamline marketing workflows.

EXPERIENCE

TRAINEE SOFTWARE ENGINEER | LIVEROOM TECHNOLOGIES PTE. LTD. | AUGUST 2024 – AUGUST 2025

- Contributed to the development and optimization of two healthcare-domain mobile applications (Health Monitoring and Secure Health Data Sharing) using Flutter.
- Enhanced application performance, stability, and consistency while managing the end-to-end mobile deployment lifecycle via App Store Connect and Google Play Console.
- Developed frontend components for an AI-Powered Blockchain Platform using Next.js.

SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, Dart, SQL.
- **Frontend:** Next.js, React, Tailwind CSS, HTML/CSS.
- **Backend & DB:** NestJS, Express.js, Node.js, PostgreSQL, Supabase, Firebase, MongoDB.
- **Mobile:** Flutter, React Native.
- **Tools/DevOps:** Git, Docker, Postman, AWS, Vercel.

EDUCATION

BENG (HONS) SOFTWARE ENGINEERING | INFORMATICS INSTITUTE OF TECHNOLOGY | 2022 - PRESENT

- Expected Graduation: September of 2026
- Relevant Modules: Object Oriented Programming; Database Systems; Software Development Group Project; Machine learning and Data Mining; Algorithms: Theory, Design, and Implementation; Industrial Placement; Final Year Project; Cyber Security; Formal Methods; Concurrent Programming; and Operational Research and Optimization.

GCE ADVANCED LEVEL | ST. THOMAS' COLLEGE, MATALE | 2018 - 2020

- Results: C, S, S (Physical Science Stream).

GCE ORDINARY LEVEL | ST. THOMAS' COLLEGE, MATALE | 2016 - 2027

- Results: 9x A's.

REFERENCE

Mr. Sahan Walpita

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Mr. Niyomal Boteju

Project Supervisor, Informatics Institute of Technology
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